

This week, lessons will cover Chapters 1 and 2 in the textbook.

## Lesson 1: Descriptive and Inferential Statistics

### Key Takeaways:

- Define what is descriptive statistics,
- Define what is inferential statistics, and
- Distinguish between descriptive and inferential statistics.

### What is “Statistics”?

**Statistics is the science of understanding data and making decisions in the face of variability and uncertainty.**

In this course we will look at:

1. Practicing Statistics: Where we analyze, understand and interpret data to make informed conclusions; and
2. Consuming Statistics: Understand and critically interpret statistical analyses reported in research articles, media, etc.

In practicing statistics and performing analysis we can provide both **Descriptive** and **Inferential Statistics**.

**Descriptive statistics**: is the analysis of your data that helps to describe, show, or summarize it in a meaningful way such that, for example, patterns might emerge from your data.

- Such statistics cannot be extended further than your specific dataset (the *sample*). They simply allow us to get a better sense of the data set that was collected.
- These statistics can be represented as **plots** (visualisations of the data using histograms, boxplots, etc.) or as **summary statistics** (numerical summaries) which include mean, median, mode (measures of central tendency), range, sample standard deviation (measures of spread) and many more.

**Inferential Statistics**: Typically, when we collect data, it is with the purpose of using it to infer conclusions around the entire population of interest (*target population*). The actual data set we collect is called a **sample**.

- Inferential statistics are techniques that allow us to use these samples to generalize about the populations from which the samples were drawn.
- The accuracy/reliability of your results will be affected by how well your sample represents the population of interest.
- Unfortunately, we rarely know anything with certainty and most conclusions will be based on a level of certainty/ confidence. More on this topic will be covered in later weeks.

### Practice

None

## Lesson 2: Parameters, Statistics and Random Sampling

### Key Takeaways:

- Understand key terminology
- Distinguish between populations and samples
- Distinguish between parameters and statistics
- Distinguish between different probability and non-probability sampling techniques
- Understand the concept of bias and be able to identify sources of bias

### Sections 2.1-2.3: Terminology and Definitions

- **Individuals/ units/ cases:** represent the objects on which measurements are taken.
- **A parameter:** describes a numerical characteristic of a population. In most cases this is unknown since we can't collect information on all the units in a population.
- **A statistic:** Is a numerical characteristic of a sample.
- **Population** (sometimes referred to as the *target population*): complete set of all objects of interest. A population can be finite or infinite in nature.
- **Study population:** All accessible units. For example, if you are going to call people to collect data the study population would be every person with a registered telephone number.
- **A sample:** the subset/group of units selected from the population on which we will collect measurements.

The sample can be **biased** for the population, that is, the sample does not accurately represent the population due to systematic differences. Biased samples introduce errors. Moving from the population to the study population introduces **study bias** if the study population is not representative of the target population. For example, if you are going to call people to collect data on their household income, people who do not own phones are more likely to be lower income, so your study population would have higher average income than the target population. **Sample bias** occurs when the sample is not representative of the study population, and depends on the sampling method used. **Measurement bias** occurs when inaccuracies in your measurements occur (people being dishonest in surveys, equipment being used not maintained properly etc.).

Sometimes the target population is the same as study population. However, in other cases, the units in the study population are different to those in the sample. Think about studies done on lab mice. In these studies, we want to generalise back to a human population, but the actual sample is made up of mice. In referring to our notion of bias, you may have come across studies justifying that mice are good representatives of the biological workings of a human and hence can be used.

### Example 1

Interest lies in estimating the average height of students attending the University of Waterloo. For this scenario we have:

- **Population:** All University of Waterloo students. The population size is typically represented by **N**.
- **Units:** Individual students
- **Sample:** A possible sample can be STAT 211 students. Notice this group is smaller than our population of interest. The size of the sample is typically represented by **n**.

- What is another possible sample? Do you think that STAT 211 students would be a good representative of our population of interest?
- Characteristic of interest: The average height, therefore:
  - Parameter: the population average. This is represented by  $E[X] = \mu$  (we will learn about this in week 5), notice that this parameter is unknown since we are not collecting information on the entire population. We use our sample to get an estimate which is defined as our statistic.
  - Statistic: the sample average height. This is represented by  $\bar{x}$  (we will learn about this in week 2).

### You try 1

A mortgage company randomly samples 100 accounts of their time-sharing customers. For each of the customers sampled, the company records the number of times the customer's payment was late.

1. Describe the population of interest.
2. Describe the sample obtained.
3. Describe a possible characteristic of interest, i.e what question might they be trying to answer?

### You try 2

Although magazines, newspapers, and books have become more available in digital format, new research indicates that printed books are making a comeback. A particular study reported that 60% of Canadians read a printed book in the past year. Suppose a sample of 1000 adults in Canada is obtained.

1. Describe the population of interest.
2. Describe the sample obtained.
3. Describe a possible characteristic of interest, i.e what question might they be trying to answer?

## Sampling Methods

How are samples selected?

There are many sampling protocols that can be adopted to select a sample. And these can be broadly divided in to two categories: **Probability vs. non-probability sampling protocols.**

Non-probabilistic based sampling protocols include:

- **Self-selection:** units volunteer to be part of the sample. This method runs the risk of producing a **biased** sample, depending on the type of study being done.
- We could end up with a **convenience sample**, simply collect measurements on units that are around us. Again, this could lead to a biased sample.
  - Example: You are interested in better understanding how University students are coping with studying remotely. You decide to run a survey on the AFM 113 piazza page to gather data from your peers.
- **Quota sampling:** Researcher takes a very tailored sample that is in proportion to some attribute in the target population. The proportions for each quota must be accurate; this is sometimes difficult or impossible to calculate and the researcher has the option to set a quota higher or lower than the proportion seen in the population.
  - Example: An interviewer might be told to go out and select 20 adult men, 20 adult women, 10 teenage girls and 10 teenage boys so that they can interview them about their television viewing.

- **Judgement sampling:** units are selected so that the samplers think that the sample will be representative of the target population.
  - Sometimes, the investigator may instead choose to approach individuals with certain characteristics. This is a technique often used by the media when looking for public opinion.

### You try 3

You are interested in the cumulative average of students in the AFM program. Give an example of how you would select a sample using each of self-selection, convenience sample, quota sampling, and judgement sampling and explain how in each case, there could be bias.

In a **probability sampling protocol**, each unit in the population has a known probability of selection and some randomisation mechanism is used to select the specific units to be included in the sample. This helps us reduce bias in our sample. Common protocols include:

- Simple Random Sampling
- Stratified random sampling and,
- Cluster sampling

### Simple random sampling (SRS):

- Is the simplest form of a probability sample. A SRS of size **n** is taken when every possible subset of **n** units in the population has the **same chance** of being in the sample.
- An implication of this is that every member of the population has the same chance of being selected.
- This selection is done without replacement (once a unit is selected they can't be selected again).
- Example: You want to select a SRS of 100 employees of Tim Hortons. You assign a number to every employee in the company database and then use a random number generator to select 100 of those numbers.

### Stratified random sampling:

- Population is initially divided into subgroups called **strata** (generated based on some characteristic of interest whose subgroups occur naturally or else are defined by the investigator). Then, an SRS is selected from each stratum.
- The strata are often subgroups of interest where the elements in the same stratum tend to be **more similar** than randomly selected elements from the whole population. As such stratification often helps to *increase precision*.
- However, it requires prior knowledge of the characteristics that will be used to generate the strata and this is not always easy to identify.
- We could sample equally from each strata or sample an amount that is proportional to the size of the strata such that larger strata have more units sampled from them.
- Example: Tim Hortons has 10,000 registered employees, 6000 are male and 4000 are female. You want to draw a sample of 1000 individuals so you sort the population into two strata based on gender. Then you randomly draw 600 men and 400 women to participate in your sample.

### You try 4

A Gallup poll interviewed 1007 randomly selected adults aged 18 and older. Gallup reported that when asked "When (if ever) the effects of global warming will begin to happen", 60% of the respondents said

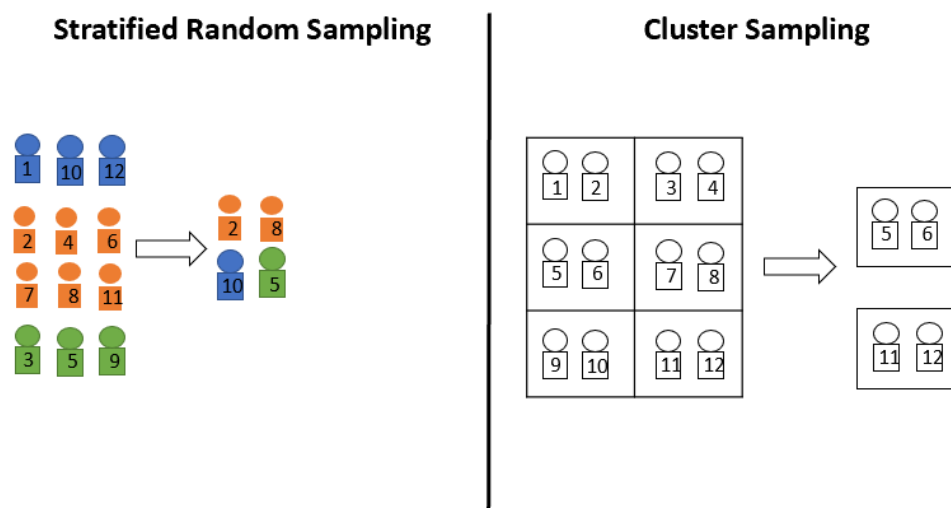
the affects had already begun. Say we were sampling adults in Canada. The sample of size 1007 adults could have been obtained in multiple ways, two of which could be:

1. By Simple Random Sampling: randomly select 1007 adults from across Canada.
2. Stratified random sampling, where we first stratify by province and then select a sample size that is proportional to the population size of the province.

Which method do you think would be better? Why?

#### Cluster sampling:

- Units in the population are “grouped” into larger sampling units, called clusters. A SRS of clusters is drawn and then a subsample of or all units in the selected clusters are used.
- Example: Hoping to learn what issues may resonate with voters in the coming election, the campaign director for a particular candidate randomly selects 10 ridings (electoral districts) in Ontario. The staff members then go and interview all the residents they can find.
- Example: The 10,000 employees working for Time Hortons are located in 500 stores across Ontario. You don’t have the capacity to travel to every store and interview the employees randomly selected. So, you randomly select 50 of the stores and interview all the employees in those stores.



In general though, what helps to ensure a sample is representative?

- Random Sampling if possible
- Sample Size: Bigger is better
- We need both! A small random sample, or a large non-random sample will not be representative. **Large random samples** are ideal.

#### Example 3

Interest lies in estimating the average height of students attending the University of Waterloo. Consider sampling first year AFM students, discuss whether such a sample would be good or bad. Explain why? What strategy may be better?

Solution:

Sampling first year AFM students may be ok from a sample size perspective, however it will likely not be representative of our population of interest. (For example, students in first year AFM are more likely to be younger on average than the average UW student).

A better strategy may be to use stratified sampling where we:

1. Stratify by Faculty
2. Randomly select a group of students from each Faculty to collectively form our sample.

We could even set-up a multi-level sampling strategy:

1. Stratify by Faculty
2. Stratify by program
3. Randomly sampling a group from each strata.

#### You try 5

A mortgage company randomly samples 100 accounts of their time-sharing customers. For each of the customers sampled the company records the number of times the customer's payment was late.

1. Describe the sampling method that the company appears to have used.
2. Discuss whether this sampling method will result in a representative sample?

#### You try 6

Although magazines, newspapers, and books have become more available in digital format, new research indicates that printed books are making a comeback. A particular study reported that 60% of Canadians read a printed book in the past year. Suppose a sample of 1000 adults in Canada is obtained.

1. Describe a possible sampling method that the company would have used.
2. Discuss whether your sampling method will result in a representative sample?

#### Practice

Chapter 2: 1, 2, 3, 4, 5

## Lesson 3: Experimental and Observational Studies

### Key Takeaways:

- Distinguish between Experiments and Observational Studies
- Identify Confounding variables

### Sections 2.4-2.5: Experiments and Observational studies

Once the sample is obtained the characteristic or characteristics of interest are collected. In most scenarios the important statistical questions involve understanding/ identifying possible relationships that exists between **multiple characteristics**. These characteristics are called **Variables (Variates)** and represent a quality/ characteristic associated with each unit. Broadly speaking these variables can be classified as one of two types:

- **Response Variable (dependent variable)**: Outcome of the study, primary variable of interest.
- **Explanatory Variable (factor/ independent variable)**: Used to explain/account for the variation in the values of the response variable.

The way we interpret our findings between the response and explanatory variable(s) is among other things dependent on the type of study that is being done.

We talk about two main types of studies

- **Observational Studies**: This is when the units of the study are disturbed as little as possible. Here we observe the units and collect information of interest on them but do not attempt to influence the responses, i.e we are analysing **pre-existing** groups. The purpose being to describe and compare pre-existing groups or situations.
- **Experiments**: Here we as the investigator actively impose some **treatment or condition** to observe the response. The purpose is to study whether the imposed treatment causes a change in the response.

### Types of Conclusions

**Observational studies** simply provide insight about the effect of changes in varying inputs. Only conclusions of **association** can be made. In other words, observational studies do not give strong evidence of a *causal* relationship. This happens since under such studies there may be other variables, that we are unaware of and hence cannot control for, that may have an effect on the response variable. These **unmeasured** variables are sometimes called **lurking** variables. Any association observed is often captured by a **correlation**, and hence we say, "*correlation does not imply causation*".

**Experimental studies** allow us to assess the effect of changing an existing process (compare a new condition to the current/old one). The units are randomly assigned to the different groups wherever possible. These groups are not pre-existing, they are created by the researchers. Conclusions of **causation** can be made. Why?

- In a well-designed randomised experiment, we don't need to worry about lurking variables. Because units are randomly allocated to each group, we can assume that there will be no systematic differences between the groups other than the conditions that were imposed by the researchers.
- Unfortunately, although experimental studies have many advantages over observational studies under many situations it may not be possible to carry out. Time, money, ethical reasons, etc. may prevent us from running an experimental study. For example:
  - In a study about the effects of smoking we can't force a person to smoke.

- In a study about shopping patterns, we can't force a person to over/under-spend.

## Confounding

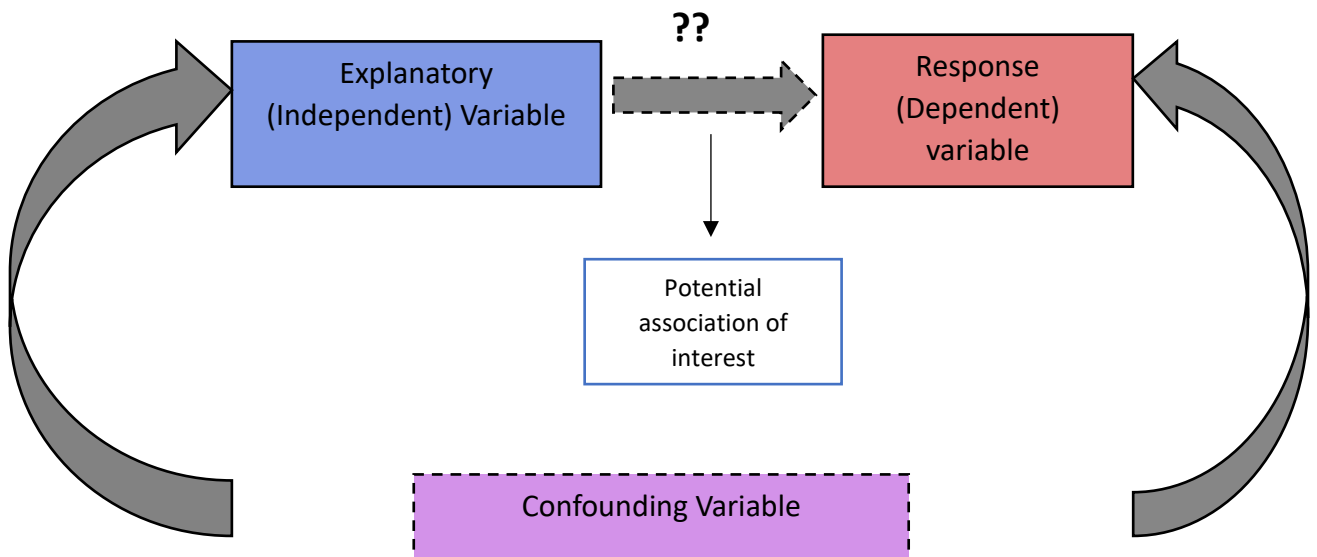
Related to the problem of interpretation associated with lurking variables is the concept of **confounding** (something that interferes with or obscures your research). **Two variables are said to be confounded if it is impossible to separate their effects on the response.**

A confounding variable may be a lurking variable (i.e. unmeasured), but it can also be a measured variable in the study. As such, confounding variables can occur in both observational and experimental studies.

Confounding between two variables occurs when the two variables vary together in the investigation.

Put another way: if there exists some other variable, not under the experimenter's control, that is associated with a factor (X) AND influences the response variable (Y) it can be difficult to know which variable is really responsible for the effect observed. Because of this we say the variable has confounded the result.

If two variables are confounded, we can't tell which is the cause of any observed change in the (average) response.



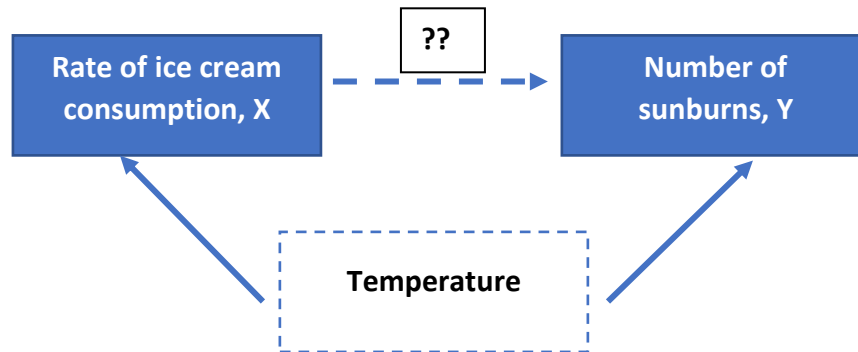
## Example 1

Researches are interested in investigating whether the rate of ice cream consumption at a local beach is associated with the number of sunburns observed over the month of July.

1. Identify the response variable of interest
2. Identify the explanatory variable of interest
3. Describe a possible confounding variable.

Solution:

1. The response variable is the “number of sunburns”.
2. The explanatory variable in the “rate of ice cream consumption”.
3. Notice that we are recording X and Y over the month of July as such a possible confounding variable in temperature. Why? If it is a very hot summer day the rate of ice cream consumption will likely be higher but so will the number of sunburns. Hence we have:



### You Try 1

Electric cars are designed to save on gasoline and help the environment. The Tesla Model S and Model 3 are designed to have extra safety features, a convenient interior design, and up to an 8-year warranty. Although an electric car has some benefits, many complain about the slow acceleration, repair expense, and overall comfort. Forty passengers are randomly selected. Each is blindfolded and taken for a ride in a traditional combustion-engine automobile and in a comparably sized electric car (over the same route). The passenger is then asked to select the car with the most comfortable ride.

1. Is this an observational study or experiment? Explain
2. What is the characteristic (variable) of interest?
3. The question mentions that drivers were always taken over the same route, why do you think this is important for the study?
4. Describe a possible confounding variable.

### Practice

Chapter 2: 6, 7, 8, 9, 10, 11, 12, 21, 22

## R Component

This week, please make sure you have R studio installed and can do basic operations in R. If you have ever used R before, this will most likely be familiar to you already so feel free to skip this portion for this week.

R may be tested so please be familiar with operations and know what the output will look like.

### Installing R studio

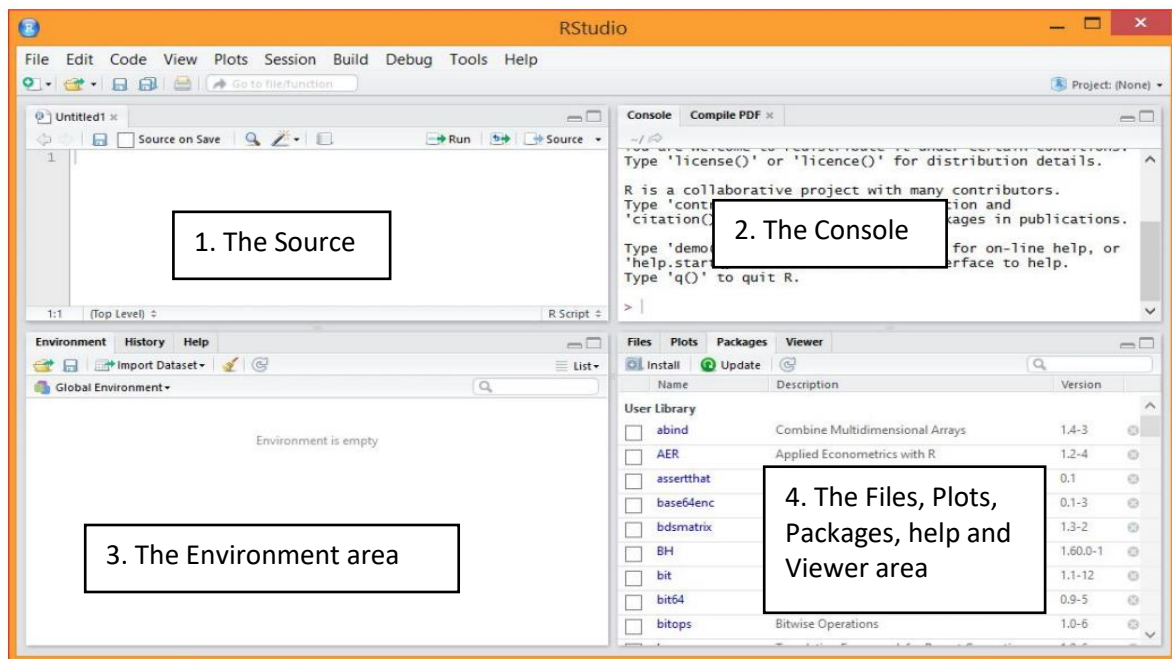
This course will use R Studio. You can download **R studio (Free version)** by following the steps outlined below:

1. Download R, available at the following link: <https://www.r-project.org/> (Note: when asked to select the CRAN, University of Toronto is the closest to us)
2. Download R studio, available at the following link: <https://www.rstudio.com/products/rstudio/download/>

### Launch R studio

After installing R studio, launch the program. Below is a description of the windows that you should see.

Note: Windows don't always appear in this order



#### 1. The Source:

This is where we write the script (code) that we want to save. You can create a new R script by selecting File > New File > R Script.

#### 2. The Console:

This is where we execute script interactively and we see the output/ results of our script.

Previously we mentioned that R is an interactive coding language. This means that every line you type is supposed to be code. In the console, when we type an instruction and hit return the line is immediately executed and the output will be shown.

### 3. The Environment area:

Shows the data set(s) currently open and the variables in each one of these data sets. The History tab keeps track of every single line of script that has been entered and run through the console.

### 4. The Files, Plots, Packages, Help, Viewer area:

Serves multiple purposes, scripts for the creation of plots will output in this window. If we need to install a specific package we search for it in the Packages tab. We can access the Help functionality if we need further information on a command etc.

## Packages

Packages are ready to use programs. Often times we may need to install packages to run a powerful analysis method and/or it can allow us to create very powerful plots etc. Currently, the CRAN package repository features over 7415 available packages: [https://cran.r-project.org/web/packages/available\\_packages\\_by\\_date.html](https://cran.r-project.org/web/packages/available_packages_by_date.html)

You will see all the available packages in R the same ones at the packages menu box. Or, you can install packages using the command

`install.packages("packagename")` . Once a package is installed, we need to load them using the command `library(packagename)` . A package is only installed once but needs to be loaded every time we need to make use of its functionality.

## Leaving Comments

It is of good form to attach comments to your code providing brief explanations around what the code is intended to do.

If you don't want to run a line in R use the hashtag/sharp/pound key #.

### Example:

```
#This is a comment and will not run!
```

## Basic operations

R can also be used as a calculator:

- Addition: `50+60`
- Subtraction: `50-60`
- Multiplication: `50*60`
- Division: `50/6`
- Raising to the power: `2^3` OR `2**3`
- Square Root: `sqrt(5)`
- Natural log: `log(5)`
- Log in another base, eg 10: `log(5, 10)`
- Exponent: `exp(10)`

## Vectors/ Datasets

If you are looking to create a variable that follows a particular sequence or repeats in nature the commands `c()`, `seq()` and `rep()` can prove to be useful.

- `c()` represents concatenate, eg `x = c(-1, 0, 1)` will create a vector containing -1,0,1
- `seq()` represents sequence
  - Create a sequence of numbers from 0 to 10 jumping in units of two (i.e `y = 0, 2, 4, 6, 8, 10`) with `y<-seq(0,10,by=2)`
- `rep()` represents repeat: `rep(a,b)` repeats element a b times

NOTE: R is caps sensitive so `x` is different from `X`.

To create a list of vectors, use the `list()` function.

We can also group variables together to create a dataset by using the `data.frame()` command.

```
x<-c(-1,0,1)
```

```
y<-c(2,4,6)
```

```
Dataset<-data.frame(x,y)
```

Vectors and other classes of objects possess certain attributes. For example, `length(x)` gives the length of the vector `x`. Vectors are a convenient way to store values of a function (e.g. a probability function or a c.d.f) or values of a random variable that have been recorded in some experiment or process.

## Reading in datasets

We can also read a table of values from a text file:

```
mydata <- read.table("c:/mydata.txt", header=TRUE)
```

Use of “header=TRUE” causes R to use the first line of the text file to get header information for the columns. If column headings are not included in the file, the argument should be omitted and we obtain a table with just the data. The R object “mydata” is a special form known as a “data frame”. Data frames that consist entirely of numeric data have a structure that is like that of numeric matrices. The names of the columns can be displayed with the command `names(mydata)`. R can also import other formats, such as csv with `dataset <- read.table("C:/filename.csv", header=T, sep=",")`.

R can make use of multiple datasets within the same script and so it is important to tell R which dataset variables being used are coming from. This can be done in a variety of ways, two ways are described below.

For example, consider a dataset of sample size  $n = 4$  with variables `X1`, `X2`, `X3` as presented below and saved in a file called ‘example.csv’

| X1 | X2 | X3 |
|----|----|----|
| 1  | 4  | 3  |
| 2  | 3  | 1  |

|   |   |   |
|---|---|---|
| 3 | 2 | 4 |
| 4 | 1 | 2 |

To start we can import our dataset: `Dataset <-read.csv("C:/example.csv", header=T)`

1. Use the `attach()` function

`attach(Dataset)` #first approach is to attach the current dataset in use

`X1` #running this will output the 4 values stored under X1

`detach(Dataset)` #once you have completed work with a dataset you can detach it from use.

2. call up variables using the `datasetname$` notation:

`Dataset$X1` #running this will again output the 4 values stored under X1 which can be found in the dataset called 'Dataset'

### Extracting elements from vectors

Try this:

- Create a vector x that contains numbers 1 to 11: `x<-seq(1,11,by=1)`
- Create another vector y that contains numbers 2 to 30: `y<- seq(2,30, by=1)`
- Join the two vectors together: `newvector<-c(x,y)`

What if we now wanted to extract certain values from `newvector`?

- Display the 22<sup>nd</sup> element, notice the use of square brackets: `newvector[22]`
- Display the fourth through to eighth elements: `newvector[4:8]`
- Negative indices can be used remove certain elements: `newvector [-2]`
  - This displays everything EXCEPT for the 2<sup>nd</sup> element
  - This does not change the original vector
  - Can avoid multiple elements: `newvector [-(3:11)]`
- Do not mix positive and negative indices. To see what happens, try `newvector [c(-2, 3) ]`, which will produce an error.
  - The problem is that it is not clear what is to be extracted: do we want the third element of `newvector` before or after removing the second one?